



Algebra 2 lecteur notes

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I-Vector spaces

I.1- Algebraic structures

Definition :

Let G be a set, we call an internal composition law on G any function from $G \times G$ to G .
Internal composition law on G is often denoted by $(*)$.

Example :

Addition (+) is an internal composition law on $\mathbb{R}$ represented as follows :

$$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$(x, y) \rightarrow x + y$$

Definition :

Let G be a set and $(*)$ be an operation on G . Then

- $*$ is to be called commutative iff :

$$\forall x, y \in G \quad x * y = y * x$$

- $*$ is to be called associative iff :

$$\forall x, y, z \in G \quad x * (y * z) = (x * y) * z$$

- $*$ has an identity element (neutral element) if :

$$\exists e \in G, \forall x \in G \quad x * e = e * x = x$$

- Each $x \in G$, an element $x' \in G$ is called the symmetric or inverse of x iff :

$$x * x' = x' * x = e$$

where $e \in G$ is the identity element.

I.1.1- Groups :

Definition :

A **group** is a set G combined with an internal composition law, denoted by $*$, such that :

- $*$ has an identity element.
- Each element in G has an inverse element in G .
- $*$ is associative.

if, in addition $*$ is commutative then $(G, *)$ is called a commutative or abelian group.



Example :

- 1- $(\mathbb{R}, +)$ is commutative group because
 - a- $\forall x, y, z \in \mathbb{R} \quad x + (y + z) = (x + y) + z$ (associative)
 - b- The neutral element for $(+)$ on $\mathbb{R}$ is 0.
 - c- Each element x of $\mathbb{R}$ has a symmetric $(-x)$ for the internal law $(+)$.
 - d- The $+$ is commutative on $\mathbb{R}$.
- 2- $(\mathbb{R}, \cdot)$ is not group because 0 does not have an inverse element.
- 3- $(\mathbb{Z}, +)$ is commutative group.

I.1.2- Ring :

Definition :

Let A be a set combined with two internal laws denoted as $(*)$ and (δ) .

We say that the triple $(A, *, \delta)$ is a **ring** if

- $(A, *)$ is commutative group.
- For all $x, y, z \in A$

$$x \delta (y * z) = (x \delta y) * (x \delta z)$$

$$\text{and } (x * y) \delta z = (x \delta z) * (y \delta z)$$

which represents left and right distributivity

- δ is associative.

If, in addition, δ is commutative, we call $(A, *, \delta)$ a commutative ring. If δ has neutral element, we refer to $(A, *, \delta)$ as a **unitary ring**

Example :

$(\mathbb{R}, +, \cdot), (\mathbb{C}, +, \cdot)$ are rings.

I.1.3- Field :

Definition :

Let $\mathbb{K}$ be a set combined with two internal laws $(*)$ and (δ) . We say that the triple $(\mathbb{K}, *, \delta)$ is a **field** if :

- $(\mathbb{K}, *, \delta)$ is unitary ring.
- $(\mathbb{K} - \{e\}, \delta)$ is group, where e is the neutral element of $(*)$.

If, in addition, δ is commutative, then we refer to $(\mathbb{K}, *, \delta)$ as **commutative field**.

Example :

$(\mathbb{K}, +, \cdot)$ is commutative field because :

- $(\mathbb{K}, +, \cdot)$ is ring.
 - 1 is neutral element of the multiplication $(\cdot)$ over $\mathbb{R}$.
- then $(\mathbb{R}, +, \cdot)$ is unitary ring
- Each element of $(\mathbb{R} - \{0\})$ has a symmetric for the law $(\cdot)$

$$\forall x \in \mathbb{R}^*, \exists y \in \mathbb{R}^* \quad x \cdot y = y \cdot x = 1 \Rightarrow y = \frac{1}{x}$$

- $(\cdot)$ is associative



$$\forall x, y, z \in \mathbb{R} \quad x.(y.z) = (x.y).z$$

I.1.4- Vector Spaces and Subspaces :

Let $\mathbb{K}$ be a commutative field (typically $\mathbb{R}$ or $\mathbb{C}$), and let E be non-empty set combined with internal law $(+)$:

$$\begin{aligned} (+): E \times E &\rightarrow E \\ (x, y) &\rightarrow x + y \end{aligned}$$

and an external law $(.)$ (called scalar multiplication (s-multiplication)) :

$$\begin{aligned} (.): \mathbb{K} \times E &\rightarrow E \\ (\lambda, x) &\rightarrow \lambda.x \end{aligned}$$

Definition :

A vector space over the field $\mathbb{K}$ (i.e $\mathbb{K}$ -vector space) is a triple $(E, +, .)$ that satisfy the following :

- 1- $(E, +)$ is commutative group.
- 2- $\forall \lambda \in \mathbb{K}, \forall u, v \in E, \quad \lambda.(u + v) = \lambda.u + \lambda.v$
- 3- $\forall \lambda, \mu \in \mathbb{K}, \forall u \in E, \quad (\lambda + \mu).u = \lambda.u + \mu.u$
- 4- $\forall \lambda, \mu \in \mathbb{K}, \forall u \in E, \quad (\lambda.\mu).u = \lambda.(\mu.u)$
- 5- $\forall u \in E, \quad 1_{\mathbb{K}}.u = u$

Remarks :

The elements of vector space are called vectors and the elements of $\mathbb{K}$ are called scalars.

If $\mathbb{K} = \mathbb{R}$ then E vector space is called « real vector space » and $\mathbb{K} = \mathbb{C}$ then E is called « complex vector space ».

Example :

$(\mathcal{A}(\mathbb{R}, \mathbb{R}), +, .)$ where $\mathcal{A}(\mathbb{R}, \mathbb{R})$ is the set of the real value function from $\mathbb{R}$ to $\mathbb{R}$ is vector space.

$(\mathbb{R}[X], +, .)$ The set of polynomials with coefficients in $\mathbb{R}$ is vector space over $\mathbb{K} = \mathbb{R}$ field.

$(\mathbb{R}^2, +, .)$ is vector space.

$(\mathbb{R}^n, +, .)$ is $\mathbb{R}$ -vector space ($n \in \mathbb{N}^*$).

Definition : (vector subspace)

Let E be a $\mathbb{K}$ -vector space and F is subset of E ($F \subset E$). We call F a vector subspace if it has the structure of vector space which means if F satisfies the axioms of vector space (in the previous Definition) with same internal and external law of E .



Theoreme :

Let E be a K -vector space and $F \subset E$. F is said to be vector subspace iff :

1. $0_E \in F$.
2. $\forall u, v \in F: u + v \in F$
3. $\forall u \in F, \forall \lambda \in \mathbb{K} : \lambda u \in F$.

Example :

The set $F = \{(x, y, z) \in \mathbb{R}^3; x = y + 2z\}$ is vector subspace.

For E vector space, $\{0_E\}$ is always vector subspace

I.1.5-Sum Vector Subspaces :**Defenition :**

Let E be a K -vector space and $F_1, F_2 \dots F_i$ ($i \in \mathbb{N}, i \geq 1$) are vector subspaces of E . The sum of $F_1, F_2 \dots F_i$ is the set of all possible elements of $F_1, F_2 \dots F_n$

$$F_1 + F_2 + \dots + F_i = \{u_1 + u_2 \dots u_i \mid u_1 \in F_1 \dots u_i \in F_i\}$$

Example : Let

$$F = \{(x, 0, 0) \in \mathbb{R}^3 \mid x \in \mathbb{R}\}$$

$$G = \{(0, 0, z) \in \mathbb{R}^3 \mid z \in \mathbb{R}\}$$

$$\text{Then } F + G = \{(x, 0, z) \in \mathbb{R}^3 \mid x, z \in \mathbb{R}\}$$

Definition : (Theorem :Direct sum)

Let F_1 and F_2 be a subspaces of vector space E . The vector space E is said to be a **direct sum** of F_1 and F_2 iff :

1. $E = F_1 + F_2$
 2. $F_1 \cap F_2 = \{0_E\}$
- and we write $E = F_1 \oplus F_2$

Example

Suppose F_1 is the subspace of $\mathbb{R}^2$ of those vector whose last coordinate equals 0, and F_2 is the subspace of $\mathbb{R}^2$ of those vectors whose first coordinate equals 0,

$$F_1 = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\} \quad F_2 = \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\}$$

1. $F_1 + F_2 = \{(x, 0) + (0, y) = (x, y) \mid x, y \in \mathbb{R}\} = \mathbb{R}^2$
2. $F_1 \cap F_2 = \{(0, 0)\}$

$$\text{Then } \mathbb{R}^2 = F_1 \oplus F_2.$$



I.2- Vector spaces Bases

I.2-1-Linear combinations and spans

Definition : (Linear combination)

A **linear combination** of list $(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$ of vectors in E is a **vector** of the form

$$\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n$$

Where $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{K}$

Example

- $(17, -4, 5)$ is linear combination of the list of vectors $((2, 1, -3), (1, -2, 4))$ because i can find $\lambda_1 = 6, \lambda_2 = 5$ where $(17, -4, 2) = 6(2, 1, -3) + 5(1, -2, 4)$.
- $(17, -4, 5)$ is not a linear combination of $((2, 1, -3), (1, -2, 4))$ because we can not find any $\lambda_1, \lambda_2 \in \mathbb{R}$ satisfy :

$$(17, -4, 5) = \lambda_1(2, 1, -3) + \lambda_2(1, -2, 4)$$

(more exactly $\begin{cases} 17 = 2\lambda_1 + \lambda_2 \\ -4 = \lambda_1 - 2\lambda_2 \\ 5 = -3\lambda_1 - 4\lambda_2 \end{cases}$ has no solutions)

Definition : (Spans)

The set of all linear combinations of the list of vectors $(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$ in E is called **span** of $\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n$ denoted **span** $(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$. In other words :

$$\text{span}(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n) = \{\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n \mid \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{K}\}$$

Example

- $(17, -4, 2) \in \text{span}((2, 1, -3), (1, -2, 4))$
 - $(17, -4, 5) \notin \text{span}((2, 1, -3), (1, -2, 4))$
- $(2, 1, -3) \in \text{span}((2, 1, -3), (1, -2, 4))$ because $((2, 1, -3) = \lambda_1(2, 1, -3) + \lambda_2(1, -2, 4)$ where $\lambda_1 = 1$ and $\lambda_2 = 0$)

Properties

Let $(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$ be list of vectors in E then

. $\forall i \in \{1, \dots, n\}, \mathbf{u}_i \in \text{span}(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$

. $\text{span}(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$ is a subspace of E

. If $F \subset E$ is subspace such that $\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n \in F$ then $\text{span}(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n) \subset F$

Definition :

If $\text{span}(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n) = E$ then we say that $(\mathbf{u}_1, \mathbf{u}_2 \dots \mathbf{u}_n)$ spans E and we call E finite dimensional space .

Example

The list of usual vectors $((1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1))$ spans $\mathbb{R}^4$



I.2-2-Linear Independence

Definition :

A list of vectors $(u_1, u_2 \dots u_n)$ is called **linearly independent** if the only solution for $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{K}$ of the equation $\lambda_1 u_1 + \lambda_2 u_2 + \dots + \lambda_n u_n = \mathbf{0}_E$ is $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$

$(u_1, u_2 \dots u_n)$ is called **linearly dependent** if there exist $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{K}$ not all zero, such that

$$\lambda_1 u_1 + \lambda_2 u_2 + \dots + \lambda_n u_n = \mathbf{0}_E.$$

Example

. The list of vectors $((1,0,0), (0,1,0), (0,0,1))$ is linearly independent in $\mathbb{R}^3$

. The list of vectors $((2,3,1), (1, -1, 2), (7,3,8))$ is linearly dependent in $\mathbb{R}^3$ (because i can find $\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = -1$ where $\lambda_1(2,3,1) + \lambda_2(1, -1, 2) + \lambda_3(7,3,8) = (0,0,0)$)

I.2-3-Bases

Definition :

A list of vectors $(u_1, u_2 \dots u_n)$ is a **basis** for the finite-dimensional vector space E if :

- 1- $(u_1, u_2 \dots u_n)$ is linearly independent.
- 2- $E = \text{span}(u_1, u_2 \dots u_n)$.

Example

- The list of vectors $((1,0,0), (0,1,0), (0,0,1))$ of $\mathbb{R}^3$ is a basis of $\mathbb{R}^3$ (is called mostly usual basis or standard basis of $\mathbb{R}^3$).
- The list of vectors $((1,2), (1,1))$ is a basis of $\mathbb{R}^2$.

Corollary : Every finite- dimensional vector space has a basis

Theorem

Every linearly independent list of vectors in a finite-dimensional vector space E can be extended to a basis of E .

I.2-4-Dimension

Definition :

We call the length of any basis of E the **dimension** of E , and we denote this by ***dim* E**

Theorem

Let E be a vector space with ***dim* $E = n$**

- If $E = \text{span}(u_1, u_2 \dots u_n)$ then $(u_1, u_2 \dots u_n)$ is basis of E .
- If $(u_1, u_2 \dots u_n)$ is linearly independent in E then $(u_1, u_2 \dots u_n)$ is basis of E .

Properties

If F and G are subspaces of finite-dimensional vector space E then :



$$\dim(F + G) = \dim(F) + \dim(G) - \dim(F \cap G)$$

I.3- Linear maps

Throughout this part E and F denoted vector spaces over $\mathbb{K}$ and $n \in \mathbb{N}^*$

Definition :

A **linear map** from E to F is a function $T: E \rightarrow F$

That satisfy :

- 1- Additivity : $T(u + v) = T(u) + T(v)$ for all $u, v \in E$
- 2- Homogeneity : $T(\lambda u) = \lambda T(u)$ for all $\lambda \in \mathbb{K}$ and $u \in E$

Remarks

The properties 1 and 2 can be written as one $T(\lambda u + v) = \lambda T(u) + T(v)$ for all $\lambda \in \mathbb{K}$, for all $u, v \in E$

- The set of all linear maps from E to F is denoted by $L(E, F)$. Moreover if $E = F$, then we write $L(E, E) = L(E)$ and T is called in $L(E)$ a **linear operator** on E

Example

- The zero map $\mathbf{0}: E \rightarrow F$ mapping every element $u \in E$ to $0 \in F$ is linear
- The identity map $\mathbf{I}: (1) E \rightarrow E$ defined as $\mathbf{I}(u) = u$ is linear
- Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 $(x, y) \rightarrow (x - 2y, 3x + y)$

Check the linearity of T :

Then for $(x, y), (x', y') \in \mathbb{R}^2$ we have :

$$\begin{aligned} T((x, y) + (x', y')) &= T(x + x', y + y') \\ &= ((x + x') - 2(y + y'), 3(x + x') + (y + y')) \\ &= (x - 2y, 3x + y) + (x' - 2y', 3x' + y') \\ &= T(x, y) + T(x', y') \end{aligned}$$

Similarly for $(x, y) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$ we have :

$$\begin{aligned} T(\lambda(x, y)) &= T(\lambda x, \lambda y) \\ &= (\lambda x - 2\lambda y, 3\lambda x + \lambda y) \\ &= \lambda(x - 2y, 3x + y) = \lambda T(x, y) \end{aligned}$$

T is linear map (operator)

Definition : (Kernel of linear map)

Let $T: E \rightarrow F$ be a linear map, then the **kernel** (or **null space**) of T is the set of all vectors in E such that : $\ker T = \{u \in E | T(u) = \mathbf{0}_F\}$

Example

We recall T from previous example

$$\ker T = \{u \in \mathbb{R}^2 | T(u) = \mathbf{0}_{\mathbb{R}^2}\} = \{(x, y) \in \mathbb{R}^2 | (x - 2y, 3x + y) = (0, 0)\}$$



$$\text{Ker}T = (0,0)$$

If $T \in L(E, F)$ where $T(u) = 0$ for every $u \in E$ then $\text{ker}T = E$

Theorem (the kernel is a subspace)

Let $T \in L(E, F)$, then $\text{ker}T$ is a subspace of E

Proof T is linear map then

$$T(0) = 0 \text{ thus } 0 \in \text{ker}T$$

Suppose $u, v \in \text{ker}T$ then

$$T(u + v) = T(u) + T(v) = 0 + 0 = 0$$

Hence $u + v \in \text{ker}T$. thus $\text{ker}T$ is closed under addition

Suppose $u \in \text{ker}T$ and $\lambda \in \mathbb{R}$ then

$$\text{thus } \lambda u \in \text{ker}T$$

Hence $\text{ker}T$ is a subspace of E

Theorem

Let $T \in L(E, F)$ then :

$$T \text{ is injective} \Leftrightarrow \text{Ker}T = 0_E$$

Theorem

Let $T \in L(E, F)$, then $\text{range}(T)$ (some mathematicians use notation $\text{Im}(T)$) is a subspace of F

Definition :

A linear map $T: E \rightarrow F$ is called surjective if $\text{range}(T) = F$

A linear map $T: E \rightarrow F$ is called bijective if T is both injective and surjective

Theorem

Suppose E is finite -dimensional vector space and $T \in L(E, F)$ then $\text{range}(T)$ is finite-dimensional and $\dim E = \dim(\text{ker}T) + \dim(\text{range}(T))$

Example

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \\ (x, y) \rightarrow (x - 2y, 3x + y)$$

Has $\text{ker}(T) = \{0_{\mathbb{R}^2}\}$ and $\text{range}(T) = \mathbb{R}^2$

$$\text{Then } \dim \mathbb{R}^2 = \dim(\text{ker}T) + \dim(\text{range}(T)) = 0 + 2 = 2$$

Corollary

Let $T \in L(E, F)$

- If $\dim(E) > \dim(F)$ then T is not injective
- If $\dim(E) < \dim(F)$ then T is not surjective.



II-Matrices

Through this chapter $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, m and n are integers in $\mathbb{N}^*$.

II-1-Matrix of linear map

Let E, F be $\mathbb{K}$ -vector spaces of dimensions n and m respectively and let $B = \{e_1, e_2, \dots, e_n\}$ be basis over E and $B' = \{e'_1, e'_2, \dots, e'_m\}$ is basis over F .

T is the linear map from E to F , where the vectors $T(e_1), T(e_2), \dots, T(e_n)$ are vectors in F and B' is basis over F (for $j = 1, \dots, n$)

$$T(e_j) = \lambda_{1j}e'_1 + \lambda_{2j}e'_2 + \dots + \lambda_{mj}e'_m$$

the table

$$\begin{pmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \\ \lambda_{11} & \lambda_{12} & \dots & \lambda_{1n} \\ \lambda_{21} & \lambda_{22} & \dots & \lambda_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m1} & \lambda_{m2} & \dots & \lambda_{mn} \end{pmatrix} \begin{pmatrix} e'_1 \\ e'_2 \\ \vdots \\ e'_m \end{pmatrix}$$

Is called matrix of linear map T related to the basis B and B' is noted mostly $Mat_{BB'}(T)$

Remarks :

1. If $E = F$, then $Mat_B(T)$.
2. The matrix $Mat_{BB'}(T)$ depends on the basis B vectors number and the basis B' .

Example :

Let T be the linear map of $\mathbb{R}^3$ to $\mathbb{R}^2$ defined as follows

$$\begin{aligned} T: \mathbb{R}^3 &\rightarrow \mathbb{R}^2 \\ (x, y, z) &\rightarrow (2x - y, x + y + z) \end{aligned}$$

Let the usual basis of $\mathbb{R}^3$: $B = \{(1,0,0), (0,1,0), (0,0,1)\}$ and $B' = \{(1,0), (0,1)\}$ the usual basis of $\mathbb{R}^2$. Then

$$T((1,0,0)) = (2,1) = 2 \times (1,0) + (0,1).$$

$$T((0,1,0)) = (-1,1) = -1 \times (1,0) + (0,1).$$

$$T((0,0,1)) = (0,1) = 0 \times (1,0) + (0,1).$$

The matrix of T

$$Mat_{BB'}(T) = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$



2. let g be a linear map over the $\mathbb{R}_3[X]$ into $\mathbb{R}_3[X]$. where $g : p \rightarrow X^2 p'' + p(1)$ with the usual basis of $\mathbb{R}_3[X]$ is given $B = \{1, X, X^2, X^3\}$

$g(1) = 1, g(X) = 1, g(X^2) = 2X^2 + 1$ and $g(X^3) = 6X^3 + 1$.
then

$$Mat_B(g) = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix}$$

II-2-Matrix

Definition : we call a matrix of **type** (m, n) (or **dimensions** $m \times n$) over $\mathbb{K}$ field any numbers of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$A = (a_{ij})$ (where $1 \leq i \leq m, 1 \leq j \leq n$) is a called a **rectangular matrix** where m the rows number and n the columns number. The numbers $a_{11}, a_{12}, \dots, a_{mn}$ are known as elements of the matrix A , In any matrix the element a_{ij} the first suffix i indicates that the element stands in the i^{th} row and the second suffix j indicates that the element stands in the j^{th} column.

Notation

- The set of matrices over K of **type** (m, n) denoted by $M_{m,n}(\mathbb{K})$, if $m = n$ the set denoted by $M_n(\mathbb{K})$.
- If $m = n$ the matrix $A = (a_{ij})$ is a called **square matrix**
- If $n = 1$ then A is called a **column matrix** (column vector) : $A = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}$
- If $m = 1$ then A is called a **row matrix** (row vector) : $A = (a_{11} \ a_{12} \ \dots \ a_{1n})$
- The Zero matrix** of the **type** (m, n) is the matrix where all a_{ij} elements are equal to 0 it can be denoted by $0_{m \times n}$

Example : $0_{3 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

II-3-Operation on Matrices

II-3-1. Equality of two matrices :

Two matrices A and B are said to be equal if

- They are of same order (type)
- Their corresponding elements are equal.

That is if $A = (a_{ij})_{m \times n}$ and $B = (b_{ij})_{m \times n}$ then $a_{ij} = b_{ij}$ for all i and j .

II-3-2. Scalar multiplication of matrix:

Let λ be a scalar then scalar product of matrix $A = (a_{ij})_{m \times n}$ given denoted by λA



$$\lambda A = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \dots & \lambda a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \dots & \lambda a_{mn} \end{pmatrix}$$

II-3-3. Addition of two matrices :

Let $A = (a_{ij})_{m \times n}$ and $B = (b_{ij})_{m \times n}$ are two matrices with same order then the sum $A + B$ is given by $A + B = (a_{ij})_{m \times n} + (b_{ij})_{m \times n} = (a_{ij} + b_{ij})_{m \times n}$

II-3-4. Multiplication of two matrices :

Two matrices are said to be **confirmable for product** if AB if number of columns in A is equal to the number of rows in matrix B . Let $C = AB$ be the matrix product of the order $m \times r$ where

$$A = (a_{ij})_{m \times n} \text{ and } B = (b_{ij})_{n \times r}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \dots + a_{in} b_{nj}$$

Theorem

Let E, F and G be vector-spaces. B_1, B_2, B_3 are basis of these vector spaces

Let $T \in L(E, F)$ and $S \in L(F, G)$

$$Mat_{B_1 B_3}(S \circ T) = Mat_{B_2 B_3}(S) Mat_{B_1 B_2}(T)$$

II-3-5. Transpose:

The transpose of matrix $A = (a_{ij})_{m \times n}$ written $A' (A' \text{ or } A^t)$ is the matrix obtained by writing the rows of A in order as columns. That

$$A = (a_{ij})_{n \times m}$$

Example (II-3) :

Let A, B :

$$A = \begin{pmatrix} 1 & 3 & -4 \\ 0 & -5 & 2 \\ 2 & 1 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 2 & 1 & 4 \\ 0 & -3 & -1 & 6 \end{pmatrix}$$

Calculate each case if it's possible $A + B, AB, BA, B', A', B + AB$.

$A + B$ is not possible because A, B are not of the same order .

$$AB = \begin{pmatrix} 1 & 3 & -4 \\ 0 & -5 & 2 \\ 2 & 1 & 3 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 2 & 1 & 4 \\ 0 & -3 & -1 & 6 \end{pmatrix}$$

AB

$$= \begin{pmatrix} 1 \times 1 + 0 \times 3 + 0 \times -4 & 0 \times 1 + 3 \times 2 + (-3) \times -4 & 1 \times 3 + 3 \times 1 + (-1) \times -4 & 1 \times 1 + 3 \times 4 + 6 \times -4 \\ 0 \times 1 + 0 \times -5 + 0 \times 2 & 0 \times 0 + (-5) \times 2 + 2 \times -3 & 0 \times 3 + (-5) \times 1 + (-1) \times 2 & 0 \times 0 + 4 \times -5 + 6 \times 2 \\ 1 \times 2 + 0 \times 1 + 0 \times 3 & 0 \times 2 + 2 \times 1 + 3 \times -3 & 2 \times 3 + 1 \times 1 + (-1) \times 3 & 2 \times 1 + 1 \times 4 + 6 \times 3 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & 18 & 10 & -11 \\ 0 & -16 & -7 & -8 \\ 2 & -7 & 4 & 24 \end{pmatrix}$$

BA it's impossible because A and B are not confirmable.



$$B' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -3 \\ 3 & 1 & -1 \\ 1 & 4 & 6 \end{pmatrix}, \quad A' = \begin{pmatrix} 1 & 0 & 2 \\ 3 & -5 & 1 \\ -4 & 2 & 3 \end{pmatrix}$$

$$B + AB = \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 2 & 1 & 4 \\ 0 & -3 & -1 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 18 & 10 & -11 \\ 0 & -16 & -7 & -8 \\ 2 & -7 & 4 & 24 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1 & 0+18 & 10+3 & 1+(-11) \\ 0+0 & -16+2 & 1+(-7) & -8+4 \\ 2+0 & -3+(-7) & 4+(-1) & 24+6 \end{pmatrix}$$

$$B + AB = \begin{pmatrix} 2 & 18 & 13 & -10 \\ 0 & -14 & -6 & -4 \\ 2 & -10 & 3 & 30 \end{pmatrix}$$

Properties of some operations

- $(A')' = A$
- $(\lambda A)' = \lambda A'$
- $(A + B)' = A' + B'$
- $(AB)' = B' A'$
- Matrix addition is commutative $A + B = B + A$
- Matrix addition is associative $A + (B + C) = (A + B) + C$
- The matrix product is distributive with respect to addition that :

$$A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

- The matrix product is associative

$$A(BC) = (AB)C = ABC$$

- $\lambda(AB) = (\lambda A)B = A(\lambda B)$
- If A and B are matrices of the same type and $AB = BA$ we say A and B commute. In this case we can define $(A + B)^P$ the binomial formula for $\forall P \in \mathbb{N}^*$

$$(A + B)^P = \sum_{k=1}^P \binom{P}{k} A^{P-k} B^k$$

II-4-Square Matrices

As we defined previously the square matrix is the matrix of type $n \times n$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

II-4-1. Integral power of matrices:

Let A be square matrix of the order n , and r be positive integer then we define

$$A^r = A \times A \times \dots \times A \text{ (} r \text{ time multiplication)}$$

$$A^2 = A \times A$$



$$A^3 = A \times A \times A$$

- The power of matrix are commutative that is

$$A^r A^s = A^{r+s} = A^s A^r$$

II-4-2. Special Types of square matrices :

Definition : (Diagonal matrix)

A square matrix is $A = (a_{ij})_{n \times n}$ is called a diagonal matrix if each of its non-diagonal element is zero (the diagonal element the elements of the indices $i = j$). Example : $A =$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Definition : (Identity matrix)

A **diagonal matrix** whose diagonal elements are equal to 1 denoted by I_n

$$\text{Example : } I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Definition : (Upper Triangular matrix)

A square matrix said to be a **Upper triangular matrix** if $a_{ij} = 0$ for $i > j$.

$$\text{Example : } A = \begin{pmatrix} 5 & 7 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Definition : (Lower Triangular matrix)

A square matrix said to be a **Lower triangular matrix** if $a_{ij} = 0$ for $i < j$.

$$\text{Example : } A = \begin{pmatrix} -1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 8 & 3 \end{pmatrix}$$

Definition : (Symmetric matrix)

A square matrix is $A = (a_{ij})_{n \times n}$ is said to be a **symmetric** if $A = A'$

$$\text{Example : } A = \begin{pmatrix} 8 & -2 & 7 \\ -2 & -9 & 3 \\ 7 & 3 & 5 \end{pmatrix}$$

A is a **symmetric** matrix.

Definition : (Skew- Symmetric matrix)

A square matrix is $A = (a_{ij})_{n \times n}$ is said to be a **skew- symmetric** if the elements a_{ij} of the Uper triangular part are equal in lower triangular part $-a_{ij}$. Example :



$$B = \begin{pmatrix} 8 & -2 & 7 \\ 2 & -9 & 3 \\ -7 & -3 & 5 \end{pmatrix}$$

B is skew-symmetric matrix.

Trace of matrix

The trace of square matrix A is the sum of the diagonal entries of A denoted $\text{Tr}(A)$.

Example

trace of B from the previous example is 4 : $\text{Tr}(B) = 8 + (-9) + 5 = 4$.

Some definitions

- A matrix A such that $A^2 = A$ is called **Idempotent**.
- If A is a matrix and r is the least positive integer such that $A^{r+1} = A$, Then A is called **periodic** of period r .
- If A matrix for which $A^r = \mathbf{0}$ the zero matrix. A is called **nilpotent**. If r is the least positive integer for which this is true. A is said to be nilpotent of order r .

II-4-3. Inverse of matrix and Elementary row Operations

II-4-3.a. Determinant

Definition : Let $A = (a_{ij})_{n \times n}$ be square matrix of order n , then the number $|A|$ called **determinant** of the matrix A .

1. Determinant of 2×2 :

$$\text{Let } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ then } |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$$

2. Determinant of 3×3 :

$$\text{Let } B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\text{Then } |B| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$|B| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Properties of Determinant

Let $A = (a_{ij})_{n \times n}$ and $B = (b_{ij})_{n \times n}$ are two matrices, $\lambda \in \mathbb{R}$

- $\det(AB) = \det(A) \det(B)$
- $\det(A') = \det(A)$
- $\det(\lambda A) = \lambda^n \det(A)$

Singular and non-singular matrix

Definition Let A be matrix of order n . A is called a singular matrix if $|A| = 0$ and non-singular otherwise.



II-4-3.b. Minor and cofactors

Definition Let $A = (a_{ij})_{n \times n}$ be square matrix of order n . Then M_{ij} denoted a sub matrix of A with order $(n - 1) \times (n - 1)$ obtained by deleting its i^{th} row and j^{th} column. The determinant $|M_{ij}|$ is called the minor of the element a_{ij} of A .

The cofactor of a_{ij} denoted by A_{ij} and is equal to $(-1)^{i+j}|M_{ij}|$

Theorem 1 (The laplace Expansion theorem)

Let $A = (a_{ij})_{n \times n}$ be square matrix of order n . $\det(A)$ is computed by

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

or

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}$$

Where $C_{ij} = (-1)^{i+j}|M_{ij}|$

Example

1. compute the determinat of A and the cofactor matrix of A

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & -2 & 4 \\ -3 & -1 & -3 \end{pmatrix}$$

With use of the first row

$$\det(A) = 1 \begin{vmatrix} -2 & 4 \\ -1 & -3 \end{vmatrix} - 0 \begin{vmatrix} 2 & 4 \\ -3 & -3 \end{vmatrix} + (-3) \begin{vmatrix} 2 & -2 \\ -3 & -1 \end{vmatrix}$$

$$\det(A) = 1((-2 \times -3) - (4 \times -1)) - 0((2 \times -3) - (4 \times -3)) + 1((2 \times -1) - (-2 \times -3))$$

$$\det(A) = 2 \neq 0$$

Or using the first column

$$\det(A) = 1 \begin{vmatrix} -2 & 4 \\ -1 & -3 \end{vmatrix} - 2 \begin{vmatrix} 0 & 1 \\ -1 & -3 \end{vmatrix} + (-3) \begin{vmatrix} 0 & 1 \\ -2 & 4 \end{vmatrix}$$

$$\det(A) = 2 \neq 0$$

A is non-singular matrix

$$\text{cofactor}(A) = \begin{pmatrix} \begin{vmatrix} -2 & 4 \\ -1 & -3 \end{vmatrix} & -\begin{vmatrix} 2 & 4 \\ -3 & -3 \end{vmatrix} & \begin{vmatrix} 2 & -2 \\ -3 & -1 \end{vmatrix} \\ -\begin{vmatrix} 0 & 1 \\ -1 & -3 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 3 & -3 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 3 & -1 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ -2 & 4 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 2 & -2 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} 10 & -6 & -8 \\ -1 & 0 & 1 \\ 2 & -2 & -2 \end{pmatrix}$$

2. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the cofactor matrix of A

$$\text{cofactor}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$



II-4-3.c. Adjoin Matrix

Definition The transpose of the matrix cofactors of the element a_{ij} of A denoted by $adj A$ is called adjoin of matrix A .

Example

find the adjoin matrix of the above Example

$$Adj(A) = cofactor A' = \begin{pmatrix} 10 & -1 & 2 \\ -6 & 0 & -2 \\ -8 & 1 & -2 \end{pmatrix}$$

II-4-3.1. Inverse of matrix

Definition If A and B are two matrices such that $AB = BA = I$ then each is said to be inverse of the other.

The inverse of A is denoted by A^{-1}

$$A^{-1}A = AA^{-1} = I$$

Theorem 2 (Existence of the inverse) :

The necessary and sufficient condition for square matrix A to have an inverse is that $|A| \neq 0$
(that A is non-singular).

Theorem 3 (uniqueness of the inverse) :

Inverse of a matrix if it exists is unique

Properties of the inverse

- $(A^{-1})^{-1} = A$
- $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}, \lambda \in \mathbb{R}^*$
- $(A^{-1})' = (A')^{-1}$
- $(AB)^{-1} = A^{-1}B^{-1}$

Definition (linear Group)

We call a linear Group of the order n over K , denoted $GL_n(\mathbb{K})$, the group of all invertible elements of $M_n(\mathbb{K})$.

Theorem 4

If A is non-singular matrix then $A^{-1} = \frac{1}{\det(A)} AdjA$



Example

The inverse of the previous A is

$$A^{-1} = \frac{1}{\det A} \text{Adj } A = \frac{1}{2} \begin{pmatrix} 10 & -1 & 2 \\ -6 & 0 & -2 \\ -8 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 5 & -\frac{1}{2} & 1 \\ -3 & 0 & -1 \\ -4 & \frac{1}{2} & 1 \end{pmatrix}$$

Let $B = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$

$$\det(B) = (1)(4) - (3)(2) = -2 \neq 0$$

B is non-singular, then

$$B^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -2 & \frac{3}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}$$

Theorem 5

Let E, F be K vector spaces of the same dimension n with respectively the basis B and B' . Let $T \in L(E, F)$, the linear map T is linear is **bijective** if and only if **$\text{Mat}_{B B'}(T)$ is non-singular.**

Theorem 6 (Similar Matrix)

Let $A, B \in M_n(\mathbb{K})$, A and B are said to be similar matrices if and only if $\exists P \in GL_n(\mathbb{K})$, $B = PAP^{-1}$

II-4-3.2. Elementary row Operations

Elementary transformations (operations)

Some operations on matrices called as elementary transformations. There are six types of elementary transformations three of them are row transformation and other three of them are column transformations there are as follows

- 1- Interchange of any two rows or columns
(for i^{th} row and j^{th} denoted by $R_i \leftrightarrow R_j$)
- 2- Multiplication of elements of any row or column by non-zero number λ .
(for i^{th} row multiplied by number $R_i \leftrightarrow \lambda R_i$)
- 3- Multiplication to elements of any row or column by a scalar λ and addition of it to the corresponding elements of any other row or column
(for j^{th} row multiplied by λ + to the corresponding elements of i^{th} row denoted by $R_i \leftrightarrow R_i + \lambda R_j$)



Definition (Equivalent matrix)

A matrix B is said to be equivalent to a matrix A if B can be obtained from A by forming finitely many successive elementary transformations on matrix A .

Denoted $A \sim B$

Example on how to compute the inverse using elementary transformations

The following method can be used with any non singular square matrix

Let the matrix $A = \begin{pmatrix} 0 & 11 \\ -2 & 3 \end{pmatrix}$

First we write the matrix in the form $(A|I)$

$$\left(\begin{array}{cc|cc} 0 & 11 & 1 & 0 \\ -2 & 3 & 0 & 1 \end{array} \right)$$

We reduce $(A|I)$ to $(I|A^{-1})$ by forming finitely many successive elementary transformations

$$\begin{aligned} R_1 &\leftrightarrow R_2 \left(\begin{array}{cc|cc} -2 & 3 & 0 & 1 \\ 0 & 11 & 1 & 0 \end{array} \right) \\ R_1 &\rightarrow -\frac{1}{2}R_1 \left(\begin{array}{cc|cc} 1 & -\frac{3}{2} & 0 & -\frac{1}{2} \\ 0 & 11 & 1 & 0 \end{array} \right) \\ R_2 &\rightarrow \frac{1}{11}R_2 \left(\begin{array}{cc|cc} 1 & -\frac{3}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{11} & 0 \end{array} \right) \\ R_1 &\rightarrow R_1 + \frac{3}{2}R_2 \left(\begin{array}{cc|cc} 1 & 0 & \frac{3}{22} & -\frac{1}{2} \\ 0 & 1 & \frac{1}{11} & 0 \end{array} \right) \\ A^{-1} &= \begin{pmatrix} \frac{3}{22} & -\frac{1}{2} \\ \frac{1}{11} & 0 \end{pmatrix} \end{aligned}$$

this method of finding the inverse is called Gauss Jordan Elimination.

II-4-4. Rank of matrix

The rank of matrix A is the dimension of the subvector space spanned by its columns.

To find the rank of matrix, we transform the matrix into its echelon form using the elementary transformations. Then, determine the rank by the non-zero rows.

Theorem

the rank of matrix A of $M_n(\mathbb{K})$ is equal n if and only if A is non-singular matrix

Example

$$\begin{aligned} \text{rank} \begin{pmatrix} 1 & 1 & 0 \\ -2 & 0 & 2 \\ 1 & 2 & 3 \end{pmatrix} &= \text{rank} \begin{pmatrix} 0 & 1 & 0 \\ -2 & 0 & 2 \\ -1 & 2 & 3 \end{pmatrix} C_1 \rightarrow C_1 - C_2 \\ &= \text{rank} \begin{pmatrix} 0 & 1 & 0 \\ -2 & 0 & 0 \\ -1 & 2 & 2 \end{pmatrix} C_3 \rightarrow C_3 + C_1 \end{aligned}$$



$$\begin{aligned}
&= \text{rank} \begin{pmatrix} 0 & 1 & 0 \\ -2 & 0 & 0 \\ 0 & 2 & 2 \end{pmatrix} R_3 \rightarrow R_3 - \frac{1}{2}R_2 \\
&= \text{rank} \begin{pmatrix} 0 & 1 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} R_3 \rightarrow R_3 - 2R_1 \\
&= \text{rank} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 2 & 2 \end{pmatrix} C_1 \leftrightarrow C_2 \\
&= 3
\end{aligned}$$



III-System of Linear Equations

III-1. Solution of the linear system $AX = b$

We now study how to find the solution of the system of m linear equations in n unknowns

Consider the system of equations in unknowns $x_1, x_2, \dots, x_n$ as

$$(S) \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

Is called system of linear equations with n unknowns $x_1, x_2, \dots, x_n$.

If the constants $b_1, b_2, \dots, b_m$ are all zero the system is said to be homogeneous type $b_j \in \mathbb{K}$

The system can be put in the matrix form as $AX = b$

Where $A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$

The matrix $A = (a_{ij})_{n \times n}$ is called coefficient matrix, the column matrix X is called the matrix of the unknowns and b is called as column matrix of constants.

III-2. Methods of solving system of linear equations

Solving the system (S) means to find the set of solution (the unknowns column matrix values)

Given any system of linear equations there are three possible outcomes:

- The system has a unique solution
- The system has Infinitely many solutions
- The system has no solutions.

III-2.1 –Method of Inversion

Consider the matrix equation $AX = b$ where $\det A \neq 0$ by multiplying A^{-1} , we have

$$\begin{aligned} A^{-1}(AX) &= A^{-1}b \\ X &= A^{-1}b \end{aligned}$$

Example

Let (S)

$$(S) \begin{cases} 3x + 3y - 2z = 5 \\ 2y + 7z = 0 \\ x + y - z = 3 \end{cases}$$

Where $A = \begin{pmatrix} 3 & 3 & -2 \\ 0 & 2 & 7 \\ 1 & 1 & -1 \end{pmatrix}$ and $b = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$. A is non singular $\det A = -2 \neq 0$



$$A^{-1} = -\frac{1}{2} \begin{pmatrix} -9 & 1 & 25 \\ 7 & -1 & -21 \\ -2 & 0 & 6 \end{pmatrix} = \begin{pmatrix} \frac{9}{2} & -\frac{1}{2} & -\frac{25}{2} \\ -\frac{7}{2} & \frac{1}{2} & \frac{21}{2} \\ 1 & 0 & -3 \end{pmatrix}$$

$$X = A^{-1}b = \begin{pmatrix} \frac{9}{2} & -\frac{1}{2} & -\frac{25}{2} \\ -\frac{7}{2} & \frac{1}{2} & \frac{21}{2} \\ 1 & 0 & -3 \end{pmatrix} \times \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -15 \\ 14 \\ -4 \end{pmatrix}$$

The system has unique solution $(x, y, z) = (-15, 14, -4)$

III-2.2- Cramer's rule method

Definition A system (S) is said to be **Cramer System** if the matrix A of the system (S) is non-singular square matrix.

Theorem(Cramer's rule)

Let (S) be a Cramer System in which A is matrix of (S) . The solution for unknown

$x_1, x_2, \dots, x_n$ is given by : $x_i = \frac{\det(A_j)}{\det(A)}$

Where A_j is the matrix A whose j^{th} column has been replaced by the constants in b

Example the same previous system

By Cramer's rule the solution is

$$x = \frac{\begin{vmatrix} 5 & 3 & -2 \\ 0 & 2 & 7 \\ 3 & 1 & -1 \end{vmatrix}}{-2} = -15, \quad y = \frac{\begin{vmatrix} 3 & 5 & -2 \\ 0 & 0 & 7 \\ 1 & 3 & -1 \end{vmatrix}}{-2} = 14, \quad z = \frac{\begin{vmatrix} 3 & 3 & 5 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{vmatrix}}{-2} = -4.$$

III-2.3 -Gaussian Elimination method (using Elementary row operation)

The Gaussian Elimination method is more general than the previous ones. The matrix A of the system does not have to be non-singular square matrix, it can be of type (m, n) (rectangle)

The Gaussian elimination is based on transforming the matrix of system A with the vector b into $(A|b)$ to some kind of **upper triangular matrix form** using the row operations

$$(A|b) = \left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_n \end{array} \right)$$

To

$$(A'|b') = \left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ 0 & a'_{22} & \cdots & b'_2 \\ \vdots & 0 & a'_{rn} & \vdots \\ 0 & \vdots & 0 & b'_n \end{array} \right)$$

Theorem

Any row operation used on matrix of the system will not change the corresponding solution of the system



Gaussian Elimination steps

1st step starting from the left, find the first non-zero column. The first pivot is obtained by taking the biggest (with absolute value) element in the first column. It's called Gauss pivot (suppose it's a_{11} as example)

2nd step use row operations on $R_1, R_2 \dots R_m$ to make the entries below the pivot position equal zero by $R_2 - \frac{a_{21}}{a_{11}} R_1, \dots, R_m - \frac{a_{m1}}{a_{11}} R_1$

3rd step Ignoring the row containing the pivot position, repeat step 1 and step 2 with the remaining rows.

Example

1- The previous system solution by Gauss method

$$(A|b) = \left(\begin{array}{ccc|c} 3 & 3 & -2 & 5 \\ 0 & 2 & 7 & 0 \\ 1 & 1 & -1 & 3 \end{array} \right) \text{ the pivot is 3 is in the first row}$$

$$R_3 \rightarrow R_3 - \frac{1}{3} R_1 \left(\begin{array}{ccc|c} 3 & 3 & -2 & 5 \\ 0 & 2 & 7 & 0 \\ 0 & 0 & \frac{-1}{3} & \frac{4}{3} \end{array} \right)$$

$$R_3 \rightarrow -3R_3 \left(\begin{array}{ccc|c} 3 & 3 & -2 & 5 \\ 0 & 2 & 7 & 0 \\ 0 & 0 & 1 & -4 \end{array} \right)$$

The system (S)

$$\begin{cases} 3x + 3y - 2z = 5 \\ 2y + 7z = 0 \\ z = -4 \end{cases}$$

We substitute $z = -4$ in 2nd equation $2y + 7(-4) = 0$ we get $y = 14$, then $x = -15$ from the 1st equation.

2- Let the system (S)

$$\begin{cases} x - y + 2z = 4 \\ x + z = 6 \\ 2x - 3y + 5z = 4 \\ 3x + 2y - z = 1 \end{cases}$$

The matrix for the Gaussian Elimination

$$(A|b) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{array} \right)$$

Use the row operation

$$\begin{aligned} R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - 2R_1 \\ R_4 &\rightarrow R_4 - 3R_1 \end{aligned} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{array} \right)$$



$$R_3 \rightarrow R_3 + R_2 \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 5 & -7 & -11 \end{array} \right)$$

The system (S) has no solution in the 3rd row $0x + 0y + 0z = -2$ which is contradiction

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